Detailed Notes: OpenShift Virtualization Overview

OpenShift Virtualization Architecture

- **Native Integration:** OpenShift Virtualization is a native feature included in all four versions of OpenShift, rather than a separate bolt-on product 1-3.
- **Type 1 Hypervisor:** By leveraging Linux KVM, OpenShift Virtualization operates as a Type 1 hypervisor that natively uses hardware virtualization 4-6.
- **Modernizing Workloads:** While Blue Yonder is initially looking at a "lift and shift" approach for VMs, OpenShift allows teams to incrementally containerize components (like web front-ends) within the same cluster later on 7-9.

Key Features and Capabilities

- **Change Block Tracking (CBT):** Red Hat is integrating native CBT into the Kubernetes stack, which allows storage vendors to perform incremental backups for both VMs and container persistent volumes without needing to do full copies and diffs 10-14.
- **Storage Offload:** To accelerate migrations from VMware, OpenShift can leverage native storage array capabilities (like XCOPY) to offload the disk conversion process directly to the storage array 15-17.
- **OVA & Ecosystem Support:** OpenShift boasts an extensive partner ecosystem, supporting Palo Alto virtual and containerized firewalls, Arista networking, and the direct import of existing OVA files hosted on an NFS share 18-24.

VM Management and Scheduling

- **Administrator Experience:** The OpenShift UI features a dedicated "Virtualization" perspective designed specifically for VM administrators to monitor resources, view top consumers, and provision VMs from templates or instance types 25-33.
- **Live Migration:** The platform supports live migrations (comparable to vMotion) with minimal packet loss, as well as cross-cluster live migrations driven by the Advanced Cluster Management (ACM) tool 34-40.
- **Workload Balancing (Descheduler):** OpenShift uses a scheduler and "descheduler" (similar to VMware DRS) to automatically rebalance VMs across nodes based on resource utilization 41-44. This can be customized or disabled for specific legacy VMs using performance profiles, taints, tolerations, and preferred/required affinity rules 45-56.

Storage and Networking

- **Storage Performance:** Administrators can hot-plug disks, CPUs, and memory 57, 58. For high-IO database workloads, Red Hat recommends utilizing the virtio interface as it is highly performant and serves as the equivalent to VMware's PVSCSI 59-66.
- **Network Configuration:** While VMware relies heavily on a GUI for network configuration, OpenShift handles VLAN mapping and networking primarily through YAML/API-driven Network Attachment Definitions 67-71. Administrators can use the UI to build out these configurations initially, and then export them to a Git repository for automated, fleet-wide policy enforcement via ACM 72-75.

Next Steps

- **Storage Offload Timeline:** Red Hat (John/Michael) will provide Blue Yonder with a timeline on when the storage offload migration capability will be fully available for Portworx and Pure Storage 76-78.

- **Performance Tuning Documentation:** Red Hat will find and share a specific blog post and documentation detailing how to tune storage configurations and increase IO reads for databases using virtio 79, 80.
- **Licensing and Pricing Follow-Up:** The teams will schedule a follow-up call to dive deeply into licensing, specifically defining what features are included with OpenShift Virtualization versus the broader OpenShift suite, and how the pricing compares to VMware 81-83.
- **Red Hat Summit Invitation:** Blue Yonder will internally discuss the invitation to attend the upcoming Red Hat user conference in Atlanta, though they noted it is highly unlikely they will get approval to attend 84-86.